

MADE BRANENDA JORDHY

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EDUCATION

Bandung Institute of Technology

B.E. in Informatics Engineering, GPA: 3.79 / 4.0

- **Relevant Coursework:** Operating System, Software Engineering, Geometric and Linear Algebra, Artificial Intelligence and Data Literacy, Computer Organization and Architecture, Algorithm and Programming, Computational Logic, Discrete Mathematics
- **Member:** HMIF ITB, IEEE ITB Student Branch

Aug 2024 – present

Bandung, West Java

TECHNICAL SKILLS

Languages: Python, Java, C, C++, Go, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Chi, sqlc, TailwindCSS, Prisma

Databases: PostgreSQL, CockroachDB, Supabase, Cloudflare D1/R2

DevOps & Cloud: Docker, Kubernetes, GitHub Actions, Cloudflare, AWS, Vercel, Azure

CERTIFICATIONS & AWARDS

Introduction to Back-End Development | Meta | [Credential](#)

Issued Feb 2026

Programming in Python | Meta | [Credential](#)

Issued Feb 2026

2nd Place Hackathon @ Hackfest 2026 Universitas Ciputra | [Find out more](#)

April 2026

EXPERIENCES

STEI ITB

Software Engineering Laboratory Assistant

Bandung, West Java

Aug 2026 – present

- Assist **IF2150 Software Engineering**, guiding student teams through the full **SDLC**, from project ideation, **requirements elicitation**, and **user stories** to construction and deployment, across **Waterfall, Iterative, Scrum, and Agile** process models.
- Mentor **analysis and design** with **UML modeling (activity, swimlane, use case, sequence, and class diagrams)**, covering **design concepts, strategies, and architectural design**, then reviewing artifacts for consistency against their implementation.
- Coach **verification & validation**, from **requirements validation** and **design evaluation** to **structured and automated testing** of the delivered build.
- Teach **software configuration management** and **Git best practices** (branching, code review, collaborative workflows), guiding students to organize engineering work as a **team** under **professional ethics**.

Inkubator IT (IIT)

DevOps Engineer (Technology Development)

Bandung, West Java

May 2026 – present

- Built and maintained **CI/CD pipelines** with **GitHub Actions and Docker** at a software house, automating build and deployment across multiple client web and mobile products with reliable, repeatable production releases.
- Deployed and operated applications on the **Cloudflare** edge stack (**Workers, Pages, D1, R2 object storage**) for fast global delivery, and wired **AWS SES** for transactional email (account verification, password resets, notifications).
- Evaluating a **K3s (lightweight Kubernetes)** setup to share infrastructure across client apps, letting each engagement start small and **scale horizontally** on demand.

Concorde Systems

Software Engineer

Bandung, West Java

May 2026 – present

- Owning the backend end to end at a **startup** building **ConcordeOS**, an AI-powered **offline-first** maintenance platform bridging enterprise CMMS (MaintainX) and field technicians in extreme industrial environments.
- Designed the **database schema (ERD)** and **REST API** from scratch, implementing type-safe endpoints with **Go (Chi, sqlc)** over **Supabase (PostgreSQL)**, plus JWT auth middleware and realtime data flows.
- Architecting an **asynchronous offline-first sync** layer that guarantees zero data loss in remote and underground sites, cutting MTTR for F&B, heavy manufacturing, and oil & gas clients.

LEADERSHIP & ACTIVITIES

The Sandbox 3.0

Director of Technology

Bandung, West Java

Nov 2025 – May 2026

- Led end-to-end engineering of a competition platform for **250+ users**, shipping the full registration and multi-stage submission pipeline (preliminary to final) with **Next.js, Prisma, and CockroachDB**.

- Built an internal admin panel for real-time financial verification and submission management, cutting organizers' manual review effort across the event cycle.

Arkavidia

Software Engineering Curriculum Developer

Bandung, West Java

May 2026 – present

- Designed a hands-on **software engineering curriculum** covering backend architecture, API integration, and production deployment for mentees.
- Led milestone-based build sessions on real features, guiding mentees to ship deployable end-to-end projects.

TEDxITB 9.0

Full-Stack Developer

Bandung, West Java

Nov 2025 – May 2026

- Shipped full-stack features for the official **TEDxITB website**, building user flows from account registration to content discovery across server and client.
- Delivered API integrations, database logic, and responsive UI for a **high-traffic public platform**, collaborating via GitHub-based workflows.

IEEE ITB Student Branch Batch 2025/2026

Full-Stack Developer

Bandung, West Java

Aug 2025 – June 2026

- Engineered the **IEEE Paper Explorer**, a data pipeline that fetches, processes, and renders live research-paper data from the **IEEE API** with dynamic content delivery.

PROJECTS

Selected work ([See all](#))

CarbonLink | NDVI, Next.js, FastAPI | 2nd Place Hackathon @ Hackfest Universitas Ciputra | [GitHub](#)

April 2026

- Built an **MRV carbon credit platform** (Next.js 14, TypeScript, Supabase) for Indonesian smallholder agroforestry, supporting polygon-based land registration and real-time IPCC Tier 1 carbon estimation.
- Engineered an AI tree-detection pipeline (**FastAPI, DeepForest**) on aerial imagery, fusing NDVI and land-area data to compute CO₂e estimates and drive a 5-stage MRV verification flow.

domTraverse | Next.js, React, Go, Docker, Azure | [GitHub](#)

May 2026

- Built a **DOM traversal visualizer** that scrapes any target URL, parses raw markup into a real tree, and animates **BFS and DFS** walks step by step with a full visit/match log.
- Implemented a **CSS-selector engine** (tag, class, id, and combinators) and real-time **Lowest Common Ancestor** computation; shipped with a **Go** backend (Next.js 16, React 19, TailwindCSS), Dockerized on **Microsoft Azure**.

VeilGuard | TypeScript, Chromium Extension API | [GitHub](#)

May 2026

- Built a **Chromium browser extension** that detects online gambling (judol) content on web pages in real time, scanning the live DOM and flagging matches automatically as pages load.
- Implemented exact **string-matching** from scratch with the **Knuth-Morris-Pratt** (failure function, $O(n+m)$) and **Boyer-Moore** (bad-character heuristic) algorithms, packaged as a production bundle (Node.js, npm).

Eigen Pustaka | Next.js (TypeScript), FastAPI (Python) | [GitHub](#)

Nov 2025

- Built a **book discovery engine** with TF-IDF + LSA text search, PCA image search, and personalized recommendations — all linear algebra hand-implemented without external ML libraries.
- Shipped a full-stack app (**Next.js + TypeScript, FastAPI + Python**) serving semantic text retrieval, image similarity, and recommendation workflows.

YAREU | Flet (Python), PostgreSQL, MVC Architecture | [GitHub](#)

Nov 2025

- Developed a **sustainability platform** unifying buy/sell, donations, and trade-in flows in a single user experience.
- Implemented modular Marketplace, Donations, and Trade-In subsystems with **MVC architecture** (Flet, Python, PostgreSQL).

RESEARCH PAPERS

Dynamic Programming for Battery Energy Storage Arbitrage Scheduling Considering Capacity Degradation Cost

June 2026

STEI ITB | [Paper](#)

- Topics: Dynamic Programming, Energy Storage, Battery Arbitrage, Degradation Cost, Optimal Scheduling, Algorithm Strategy.

Efficient Transformer Compression in Pre-trained Language Models Through Selective Tensor Rank Reduction

Dec 2025

STEI ITB | [Paper](#)

- Topics: Model Compression, Natural Language Processing, Transformer Efficiency, Tensor Decomposition, Low Rank Approximation, Geometric and Linear Algebra.

Anomaly Detection in AI Knowledge Graphs: A Discrete Mathematical Framework Against Graph Poisoning

June 2025

STEI ITB | [Paper](#)

- Topics: Knowledge Graphs, AI Security, Discrete Mathematics, Graph Theory, Anomaly Detection.